The UK Field Guide

A vision-grounded wildlife assistant that turns a photograph into a cited, actionable answer

1. General Overview

The UK Field Guide is a naturalist assistant that closes the gap between species identification and actionable advice. Existing identification apps answer ‘what is this?’ well, but almost none answer the question that matters next: ‘what should I do about it?’. This project takes a photograph or a written question from the user and returns an answer that is identified, explained, sourced, and safety-checked, rather than a bare species label.

The system was built for three audiences: members of the public identifying a plant, insect, fungus, or bird on a walk; local councils and conservation groups needing an invasive-status check; and agricultural bodies or land managers assessing risk. It is scoped to the United Kingdom and built around a retrieval-augmented, agentic architecture with a genuine multimodal (photo) input path.

Knowledge base at a glance

362 Source documents	**245** Species accounts	**47** Books	**27** Papers

The corpus was assembled from iNaturalist (vision identification), Wikipedia (encyclopedic species information), scientific literature via Europe PMC, PubMed Central, arXiv, and Open Library, and official UK Government invasive, protected-species, and conservation lists. Extraction was scripted end to end: legal and scientific sources were pulled with citation metadata attached at download time, so downstream ingestion never has to guess a source's provenance from its filename.

Delivered outputs include the extraction and ingestion pipeline (source download, PDF/text organisation, chunking, embedding, and loading into ChromaDB), the two-agent RAG application itself, a 30-question benchmark set spanning four real conversations, and an observability layer instrumented through LangSmith.

2. How It Works

The user opens a clean, outdoor-friendly interface with three inputs: a photo upload field, a free-text question box, and an optional region selector. Either a photo or a question is required to trigger the app; either path converges on the same agentic core.

The pipeline, step by step

- **Vision or text input.** A photograph is passed to the iNaturalist computer-vision API, which returns the top-3 species candidates with confidence scores — an honest, probabilistic starting point rather than one overconfident guess. Only the top result is carried forward into the final answer. If no photo is given, a direct text question (e.g. "Tell me about Japanese knotweed") is used to identify the species or topic instead, skipping vision entirely.
- **Wildlife Agent (Agent 1) orchestrates.** This LLM-powered agent receives the vision result or the parsed question, decides which tools to call, and builds a self-contained query. Its tools are retrieve_documents (must be called before any species-specific answer), summarise_topic, compare_species, and consult_safety_analyst.
- **Species RAG grounds the answer.** The agent queries a local ChromaDB vector store built from the 362-document corpus. Documents are whitespace-cleaned and de-duplicated by hash, then split with LangChain's fixed-character CharacterTextSplitter (1000-character chunks, 100-character overlap) so that facts are rarely cut mid-sentence. Chunks are embedded with OpenAI's text-embedding-3-small and stored using Euclidean distance.
- **Retrieve and rerank.** For each query, the top 15 chunks are retrieved by embedding similarity, then re-scored by a cross-encoder reranker (ms-marco-MultiBERT-L-12 via FlashrankRerank). A 0.7 score threshold discards weak matches, and the best 7 passages are passed to the agent — a cost/accuracy trade-off that keeps retrieval fast while pruning low-confidence context.
- **Safety Analyst (Agent 2) checks status.** Whenever the conversation touches legal protection, invasive status, or conservation status, Agent 1 delegates to a dedicated Safety Analyst agent (an agent-as-tool call). It checks a reference table and the invasive/protected lists database, keeps its own answers to a few sentences, and states plainly when the retrieved context does not support a conclusion.
- **Final, cited answer.** The Wildlife Agent synthesises the retrieved passages and the safety check into one streamed, Markdown answer: actionable advice, inline citations, and a numbered source list. Safety warnings are appended in code, not left to model judgement, so an invasive or protected-species flag can never be silently dropped by the LLM.
- **Memory and observability.** Conversation memory uses a hybrid window: the last 3 messages are kept verbatim, and everything older is folded one message at a time into a rolling summary, keeping the context size roughly flat as a chat grows. Every tool call and agent step is traced to LangSmith for latency, token, and cost monitoring.

3. Why It's Useful

The core value of the system is that it goes one step past identification into decision support, and it does so with visible honesty about its own uncertainty.

What makes it trustworthy, not just accurate

- **Uncertainty is shown, not hidden.** Presenting the top-3 candidates and their confidence scores, rather than a single label, matches how a careful naturalist would actually communicate — useful for lookalike species where a wrong guess could carry real consequences (e.g. a toxic mushroom mistaken for an edible one).
- **Every claim is grounded and cited.** Because the Wildlife Agent must call retrieve_documents before answering a species-specific question, and the final response ends with a numbered source list, users and organisations can verify a claim rather than trust an LLM's memory.
- **Safety is enforced in code.** Invasive-species alerts and legal-protection warnings are appended by the application logic, not generated by the model. This closes off the single most damaging failure mode of a naturalist assistant: an LLM quietly omitting a safety warning.

Who benefits

- Members of the public get an answer that tells them not just what a species is, but what to do next — whether to report it, avoid it, or leave it alone.
- Local councils and conservation groups get a fast, citable first-pass invasive-status check that can be used to triage reports before a specialist is involved.
- Agricultural bodies and land managers get a low-cost way to screen for regulated or hazardous species across a property.

Benchmarking discipline

The project was evaluated on a 30-question set spanning four real conversations (American Bullfrog, Small Emperor Moth, Swedish Whitebeam, and a general Dogs vs Cats comparison), run once under a baseline configuration and once under an ‘improved’ configuration (a stronger LLM, a lower reranker threshold, and wider retrieval k-values). The corrected results showed both configurations at 0% error, with the improved bundle 3.85× more expensive and roughly 2.7× slower at the median (7.06s vs 2.60s) with no measurable accuracy gain — a genuinely useful negative result that stopped an expensive upgrade from shipping on unverified assumptions.

4. Future Improvements

The current build is a working end-to-end pipeline, but several honest gaps remain, each with a concrete next step.

Known limitations → planned fixes

- **Naive chunking.** Fixed-character splitting cuts text at a set length rather than at sentence or paragraph boundaries. Moving to a semantic or structure-aware splitter would keep species facts from being severed mid-sentence.
- **Thin UK Government source coverage.** Gov.uk invasive- and conservation-status documents are currently limited. Completing ingestion of the missing lists, and growing the regional-status document count, would materially improve the Safety Analyst's coverage.
- **Unverified retrieval quality.** The semantic ranking and the FlashRank reranker's real precision on this specific corpus have not been checked by a domain specialist. A manual or expert-reviewed relevance audit is a natural next step.
- **Settings changed all at once.** The baseline-vs-improved comparison varied the LLM, reranker threshold, and both retrieval k-values simultaneously, so the test can only show whether the bundle works as a whole, not which individual change drove any effect. Future evaluation should isolate one setting at a time and re-run across multiple passes for consistency.
- **Single region, no lookalike warnings.** The system is scoped to the UK only, and identification currently returns species suggestions without seasonal or lookalike-species context (e.g. "often confused with X; check the stem"). Adding this would directly reduce the risk of a confident but wrong identification.

Stretch goals

- An offline-friendly mode using a locally run vision model, reducing dependence on the iNaturalist API.
- A report-an-invasive flow that drafts the official reporting submission for the user's region.
- A text-to-video pipeline that generates a short (45–60 second) informative video about the identified species.
- Expanding the photograph evaluation set to at least 40 labelled images, reporting both top-1 and top-3 identification accuracy alongside status-check correctness.